

Interest rate derivatives (Sensitivities)

Contents



- ▶ Why Sensitivity is important ?
- ▶ Taylor Series recap
- ▶ The case of ZCB
- ▶ The case of CB
- ▶

Why is sensitivity needed ?

► The price of a financial instrument is dependent on several risk factors. The sensitivity refers to the rate of change of price w.r.t the underlying risk factors. Sensitivity reflects exposure or risk to the respective underliers. Specifically, sensitivity is helpful for

- Local revaluation ✓
- Risk Modeling (e.g., delta VaR, FRTB)
- Hedging (e.g., delta hedging, gamma hedging)
- PnL attribution

Taylor Series

Taylor Series (Recap)

- Taylor series expansion relates change in dependent variable (y) to change in independent variable (x). This tool will be extremely valuable to relate PnL of an instrument or even a portfolio to the changes in risk factors through the sensitivities. The second order approximation is given below

• Univariate Case $y = f(x)$

$\Delta y \leftrightarrow \Delta x$

$$\Delta y \approx \underbrace{\left(\frac{dy}{dx}\right)(\Delta x)}_{\text{(Linear)}} + \frac{1}{2} \underbrace{\left(\frac{d^2y}{dx^2}\right)(\Delta x)^2}_{\text{(quadratic)}}$$

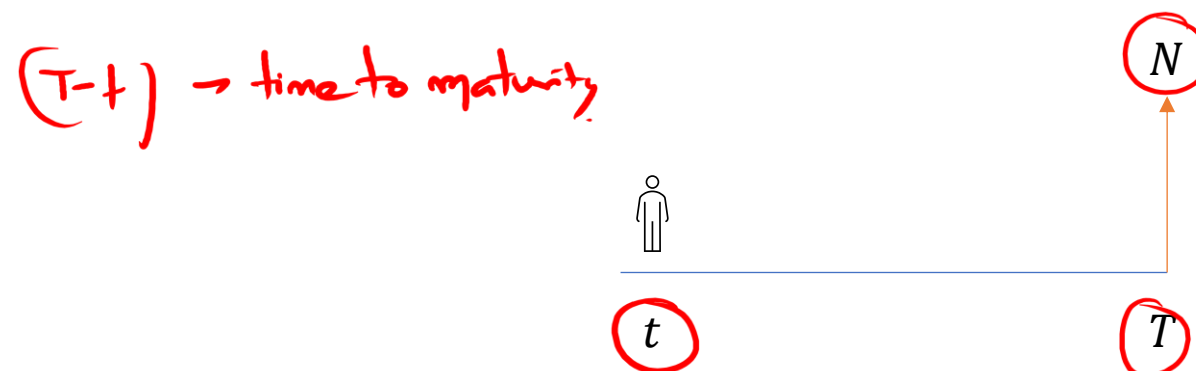
• Bivariate Case $y = f(x_1, x_2)$

$$\Delta y \approx \underbrace{\left(\frac{\partial y}{\partial x_1}\right)(\Delta x_1)}_{\text{(Linear)}} + \underbrace{\frac{\partial y}{\partial x_2}(\Delta x_2)}_{\text{(Linear)}} + \frac{1}{2} \underbrace{\frac{\partial^2 y}{\partial x_1^2}(\Delta x_1)^2}_{\text{(Quadratic)}} + \frac{1}{2} \underbrace{\frac{\partial^2 y}{\partial x_2^2}(\Delta x_2)^2}_{\text{(Quadratic)}} + \underbrace{\frac{\partial^2 y}{\partial x_1 \partial x_2}(\Delta x_1)(\Delta x_2)}_{\text{(Cross Term)}}$$

#1 – Zero Coupon Bond (cont. compounded yield)



- ▶ Let's apply Taylor Series approximation for a zero-coupon bond at time t , maturing at T and having a continuously compounded yield y .



- ▶ Value of the ZCB : $V = Ne^{-y(T-t)}$

$$\frac{\Delta V}{V} \approx -(T-t)\Delta y + \frac{1}{2}(T-t)^2(\Delta y)^2$$



#1 – Zero Coupon Bond (compounding freq. = m)

- ▶ How does the formula change when the yield is in terms of discrete compounding (freq. = m).
- ▶ Value of the ZCB :

$$V = \frac{N}{\left(1 + \frac{y}{m}\right)^{m(T-t)}}$$

$$\frac{\Delta V}{V} \approx - \frac{(T-t)}{\left(1 + \frac{y}{m}\right)} \Delta y \quad (\text{first order approx.})$$

$m \rightarrow \infty$

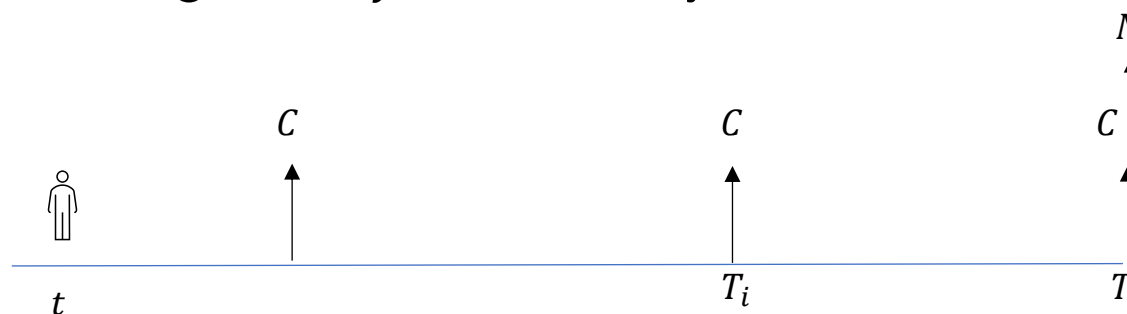
$(T-t)$

$m=1$

- ▶ Note that when $m \rightarrow \infty$, the formula is same as in case of continuous compounding.

#2 – Coupon Bond (Annual Freq.)

- We can perform the same steps for a coupon bond. Assume the coupon payments happen once a year. The yield of the bond is given as y and maturity T



Value of the CB :
$$\underline{V} = \frac{C}{(1+y)^1} + \frac{C}{(1+y)^2} + \dots + \frac{C+N}{(1+y)^T}$$

$$\Delta V \approx -\frac{1}{1+y} \left(\underline{1} \cdot \frac{C}{(1+y)^1} + \underline{2} \cdot \frac{C}{(1+y)^2} + \dots + T \cdot \frac{C+N}{(1+y)^T} \right) (\Delta y) \quad (\text{first order approx.})$$

$$\left[\frac{\Delta V}{V} \approx -\frac{\underline{D}}{1+y} (\Delta y) \right] \quad \underline{D} = \frac{\left(\underline{1} \cdot \frac{C}{(1+y)^1} + \underline{2} \cdot \frac{C}{(1+y)^2} + \dots + \underline{T} \cdot \frac{C+N}{(1+y)^T} \right)}{V}$$

Macauly Duration Vs Modified Duration

- The Macauly Duration has the interpretation of average waiting time to receive all the cashflows. The first order sensitivity is a scaled version of Macauly Duration and is called Modified Duration.

$$\frac{\Delta V}{V} \approx -\frac{(D)}{1+y}(\Delta y) \quad \rightarrow \quad \frac{\Delta V}{V} \approx -D^*(\Delta y)$$

$$\text{Modified Duration } (D^*) = \text{Macauly Duration } (D) / (1+y)$$

- Note that the relation is dependent on frequency of compounding

- Continuous compounding : $D^* = D$
- Annual Compounding : $D^* = \frac{D}{1+y}$
- 'm' times per year : $D^* = \frac{D}{1+\frac{y}{m}}$

Other forms of sensitivities

- ▶ The Duration measure links the % change in price to the change in yield. We can get a delta measure (aka money duration or \$ duration) which measures the absolute change in price. This is simply achieved by multiplying price.

$$\text{Money Duration} = \text{Modified Duration} \times V$$

To summarize

$$\frac{\Delta V}{V} \approx -(\text{Modified Duration}) \cdot (\Delta y) \quad \Delta V \approx -(\text{Money Duration}) \cdot (\Delta y)$$

$\uparrow$ $\uparrow$
\$ Duration

- ▶ Often, we use the measures DV01 or PV01 which measure the dollar sensitivity w.r.t 1 bps change in the yield. There are differences in adoption of this quantity in literature in terms of the bps change (1 or 100).

$$DV01 = -\$97$$

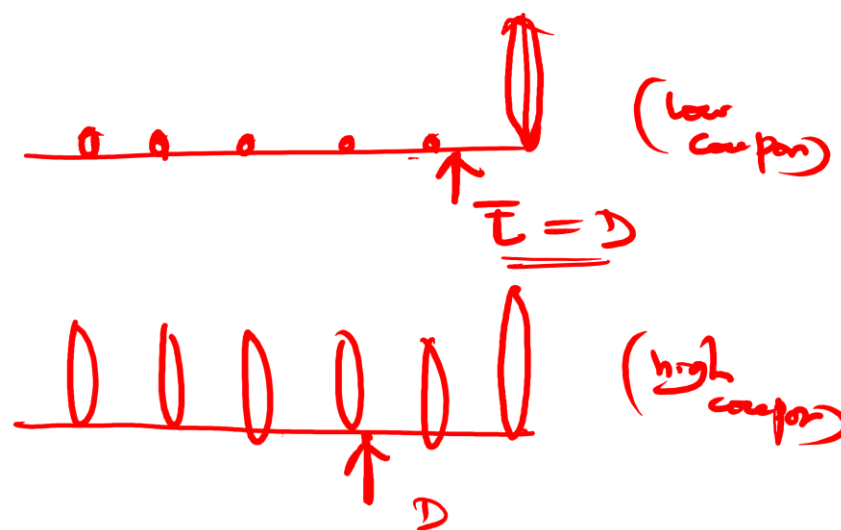
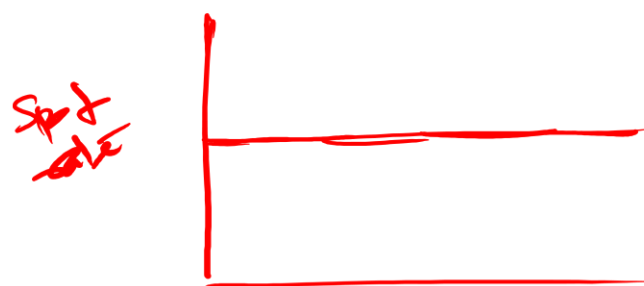
$$DV01 = -(\text{Money Duration}) \cdot (0.0001)$$

$\uparrow$

Concept Checker



- ▶ Duration of a ZCB ? $\leftarrow$
- ▶ Which is larger , Duration of a low coupon Bond or High coupon bond (everything else same) ?
- ▶ Does Macaulay duration assume a flat yield curve ? $\times$
- ▶ Duration of a consol ?
 $\quad \quad \quad = \text{perpetuity}$



Curve Duration

- ▶ The yield Duration and convexity measure the change of the bond price with respect to yield i.e., a summary statistic. It does not tell us about the drivers that result in change of yield. Many different shapes of the spot curve can lead to the same yield. So, for more pragmatic applications, we might want to pivot to sensitivity w.r.t the spot curve.
- ▶ So, **spot duration** is given as



$$D^{\text{spot}} = \frac{\left(1 \cdot \frac{C}{(1+z_1)^1} + 2 \cdot \frac{C}{(1+z_2)^2} + \dots + T \cdot \frac{C+N}{(1+z_T)^T} \right)}{V}$$

where $(z_1, z_2, \dots, z_T)$ are the zero-coupon yields (=spot rates)

- ▶ Spot duration can be used to measure price impact for parallel shifts of the spot curve.

Effective Duration & Convexity

- For bonds with embedded optionality, the duration calculation is not straightforward. We resort to numerical techniques (finite difference) to find the duration. This is called effective duration and is the most general adoption of sensitivity formula

$$y = f(x)$$

$$\frac{\Delta y}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\frac{f(x) - f(x - \Delta x)}{\Delta x}$$

$$\frac{f(x + \Delta x) - f(x - \Delta x)}{2\Delta x}$$

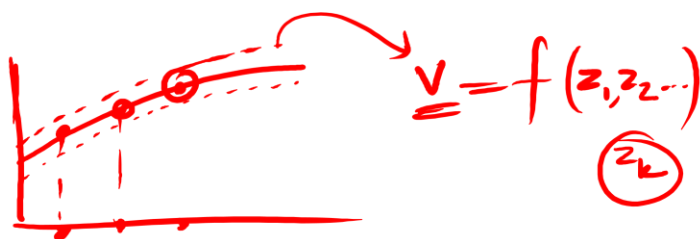
$$\text{Effective Duration} = \frac{V(\text{down}) - V(\text{up})}{2 * \Delta \text{curve} * V(\text{base})}$$

$$\text{Effective Duration} = \frac{V(\text{down}) + V(\text{up}) - 2V(\text{base})}{(\Delta \text{curve})^2 * V(\text{base})}$$

$$\left(\frac{1}{V} \frac{\partial V}{\partial y} \right) \rightarrow \text{Duration}$$

Key Rate Duration

- ▶ When viewing bond price as a function of several spot rates, a more useful measure of sensitivity is the impact of change in one key rate while keeping all other rates constant. This is called **key-rate duration**.



$$KRD(k) = -\frac{1}{V} \frac{\partial V}{\partial z_k}$$

$$V = \frac{C}{(1+z_1)} + \frac{C}{(1+z_2)^2} + \dots + \frac{C+N}{(1+z_n)^n}$$

$$\Delta V = -\frac{k}{1+z_k} \times (V_k) \cdot \Delta z_k$$

- ▶ Numerically, it can be found by shocking one of the rate while keeping others constant.

$$KRD = \frac{V(down) - V(up)}{2 * \Delta(key\ rate) * V(base)}$$

$$KRD^k = -\frac{\frac{k}{1+z_k} \cdot V_k}{V}$$

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- The top part of the image shows a complex system on the left, represented by a stick figure, a box, and various springs and forces. This is equated to a sequence of simpler systems on the right, each with a stick figure and a spring. The first system has a spring with force T_{β} . The second system has a spring with force T_{i-1} and a stick figure at time t . The third system has a spring with force $NL(T_{i-1}, T_i)\tau$ and a stick figure at time T_i .
- The bottom part of the image shows a sequence of triangles, each with a peak labeled $6m$. To the right of these triangles is the expression $D \in (0, 6m)$.

Duration of a Floater

- ▶ When the projection curve and the discounting curve are different, then we can calculate two types of durations
 - Rate duration : (because of change in benchmark rate)
 - Spread duration : (because of change in spread used in discounting term to reflect CCR)
- ▶ The durations can be calculated using finite difference

$$\left[\begin{pmatrix} \frac{\partial V}{\partial r_1} & \frac{\partial V}{\partial r_2} \end{pmatrix} \right] \rightarrow \text{Cross gamma}$$

$\uparrow$ $\uparrow$
 $\frac{\partial V}{\partial r_1}$ $\frac{\partial V}{\partial r_2}$

$$\text{Rate Duration} = \frac{V(\text{down}) - V(\text{up})}{2 * \Delta \text{rate curve} * V(\text{base})}$$

$$\text{Spread Duration} = \frac{V(\text{down}) - V(\text{up})}{2 * \Delta \text{spread curve} * V(\text{base})}$$